

MINNEAPOLIS FLYING CLOUD, MN, USA

WMO: 726579

Lat: 44.832N

Lon: 93.471W

Elev: 277

StdP: 98.05

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94963

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-21.3	-27.9	0.3	-22.9	-26.0	0.4	-20.8	10.5	-7.0	9.2	-8.0	2.9	240	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.5	23.4	31.1	22.6	29.2	21.5	25.2	30.6	24.0	28.9	22.9	27.8	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	28.8	22.4	17.7	27.6	21.3	16.5	26.3	78.5	30.3	73.5	29.0	69.2	27.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.6	7.8	-27.8	35.6	3.1	1.6	-30.0	36.7	-31.8	37.7	-33.5	38.6	-35.8	39.7		
			-28.0	27.0	3.0	1.3	-30.2	27.9	-32.0	28.6	-33.7	29.3	-35.9	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-8.7	-6.8	0.5	8.7	15.4	20.6	23.6	21.9	17.9	9.9	2.1	-5.7
	DBStd	12.52	7.07	6.76	6.90	5.67	4.77	3.66	2.97	2.96	4.63	5.20	5.92	6.61
	HDD10.0	2253	579	472	304	90	8	0	0	0	2	67	244	487
	HDD18.3	4093	837	705	554	292	117	19	2	4	63	268	486	746
	CDD10.0	1651	0	0	9	50	175	317	421	369	239	63	8	0
	CDD18.3	449	0	0	1	2	26	86	164	115	51	5	0	0
	CDH23.3	3808	0	0	3	32	258	707	1479	880	404	44	1	0
	CDH26.7	1219	0	0	0	8	86	218	524	254	119	8	0	0
Wind	WSAvg	3.8	3.8	3.9	4.0	4.4	4.0	3.6	3.3	3.1	3.6	3.9	3.9	3.7
Precipitation	PrecAvg	808	18	20	40	81	103	122	98	111	86	62	36	28
	PrecMax	1017	39	40	77	170	228	231	173	203	194	146	110	68
	PrecMin	591	3	7	19	38	18	47	46	18	12	16	3	5
	PrecStd	124	9	9	15	34	50	48	43	58	55	38	28	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	11.1	21.6	27.3	32.7	33.6	35.2	33.7	32.5	27.7	20.1	11.0
		MCWB	3.4	7.9	14.7	16.7	20.3	23.4	25.3	24.4	22.6	18.5	13.1	8.5
	2%	DB	3.9	7.0	17.1	22.8	28.9	31.2	32.7	31.2	30.0	23.9	16.3	7.1
		MCWB	1.8	4.0	12.1	13.4	18.5	21.7	24.3	23.0	21.7	16.3	11.1	4.3
	5%	DB	2.3	4.0	13.6	20.7	26.2	29.0	31.2	29.1	27.6	21.0	12.9	4.1
		MCWB	0.9	2.0	9.5	12.2	17.1	20.8	23.5	22.0	20.0	14.9	8.9	2.2
	10%	DB	0.9	2.4	10.1	17.7	23.4	27.4	29.4	27.7	25.6	17.8	10.6	2.5
		MCWB	-0.3	0.6	6.7	10.6	15.7	19.7	22.1	20.8	19.1	13.1	7.0	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	7.9	16.3	17.7	22.6	25.2	27.5	25.8	24.5	20.6	14.8	9.1
		MCDB	6.2	10.9	20.1	25.2	30.4	31.1	31.8	31.4	30.1	24.7	18.7	10.6
	2%	WB	2.1	4.2	12.9	14.9	20.1	23.4	25.8	24.4	22.8	17.6	12.0	4.6
		MCDB	3.7	6.3	16.3	21.1	26.2	28.9	31.1	29.2	27.6	21.1	15.6	6.6
	5%	WB	0.8	2.1	9.8	13.0	18.5	22.0	24.6	23.3	21.4	15.7	9.3	2.6
		MCDB	2.1	4.0	13.5	19.0	24.0	27.5	29.6	27.8	26.0	20.2	12.7	4.2
	10%	WB	-0.5	0.7	6.7	11.4	17.0	20.7	23.3	22.2	20.1	13.6	7.3	1.1
		MCDB	0.6	2.2	10.0	16.9	22.1	25.5	28.2	26.5	24.5	17.6	10.1	2.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	8.9	10.1	10.5	9.9	10.1	9.9	10.5	9.4	8.0	7.4
		MCDBR	8.9	9.1	12.7	14.5	14.5	12.0	11.4	11.2	12.6	13.7	11.9	9.2
		MCWBR	7.4	7.2	8.2	7.9	6.7	5.7	5.3	5.3	6.0	7.5	7.8	7.4
	5% WB	MCDBR	8.3	8.7	11.7	12.8	12.5	10.5	10.5	10.1	10.9	11.2	11.2	8.6
		MCWBR	7.1	6.9	8.1	7.7	6.7	6.0	5.7	5.7	6.2	7.1	7.8	7.1
Clear-Sky Solar Irradiance	taub	0.270	0.285	0.319	0.374	0.404	0.400	0.402	0.405	0.378	0.322	0.292	0.268	
	taud	2.421	2.362	2.390	2.300	2.270	2.327	2.338	2.328	2.366	2.520	2.509	2.440	
	Ebn at Noon	859	906	916	886	866	869	862	846	840	855	828	824	
	Edh at Noon	71	91	103	123	131	124	122	119	105	78	66	64	
All-Sky Solar Radiation	RadAvg	1.73	2.82	3.75	4.58	5.33	5.94	6.31	5.27	4.12	2.62	1.67	1.22	
	RadStd	0.21	0.29	0.36	0.47	0.44	0.48	0.24	0.38	0.30	0.35	0.16	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.49	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page